

ESP32-S3-LCD-2

From Waveshare Wiki

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Overview

[[Expand](#)]

Usage Instructions

ESP32-S3-LCD-2 currently provides two development tools and frameworks, **Arduino IDE** and **ESP-IDF**, providing flexible development options, you can choose the right development tool according to your project

ESP32-S3-LCD-2



(<https://www.waveshare.com/ESP32-S3-LCD-2.htm>)

2inch, 240×320, SPI

needs and personal habits.

Development Tools



(/wiki/File:180px-Arduino-IDE-logo.jpg)

Arduino IDE

Arduino IDE is an open source electronic prototyping platform, convenient and flexible, easy to get started. After a simple learning, you can start to develop quickly. At the same time, Arduino has a large global user community, providing an abundance of open source code, project examples and tutorials, as well as rich library resources, encapsulating complex functions, allowing developers to quickly implement various functions.

ESP-IDF



(/wiki/File:180px-ESP-IDF-logo.jpg)

ESP-IDF, or full name Espressif IDE, is a professional development framework introduced by Espressif Technology for the ESP series chips. It is developed using the C language, including a compiler, debugger, and flashing tool, etc., and can be developed via the command lines or through an integrated development environment (such as Visual Studio Code with the Espressif IDF plugin). The plugin offers features such as code navigation, project management, and debugging, etc..

Each of these two development approaches has its own advantages, and developers can choose according to their needs and skill levels. Arduino are suitable for beginners and non-professionals because they are easy to learn and quick to get started. ESP-IDF is a better choice for developers with a professional background or high performance requirements, as it provides more advanced development tools and greater control capabilities for the development of complex projects.

Components Preparation

- ESP32-S3-LCD-2 x1
- OV5640 camera x1
- TF card x 1 (32GB and below)
- Card reader x1

Before operating, it is recommended to browse the table of contents to quickly understand the document structure. For smooth operation, please read the FAQ carefully

Working with Arduino

This chapter introduces setting up the Arduino environment, including the Arduino IDE, management of ESP32 boards, installation of related libraries, program compilation and downloading, as well as testing demos. It aims to help users master the development board and facilitate secondary development.

Environment Setup

Download and Install Arduino IDE

- Click to visit the Arduino official website (<https://www.arduino.cc/en/software>), select the corresponding system and system bit to download

 **Arduino IDE 2.3.3**

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger.

For more details, please refer to the [Arduino IDE 2.0 documentation](#).

Nightly builds with the latest bugfixes are available through the section below.

SOURCE CODE

The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

DOWNLOAD OPTIONS

- Windows** Win 10 and newer, 64 bits
- Windows** MSI installer
- Windows** ZIP file
- Linux** Applimage 64 bits (X86-64)
- Linux** ZIP file 64 bits (X86-64)
- macOS** Intel, 10.15: "Catalina" or newer, 64 bits
- macOS** Apple Silicon, 11: "Big Sur" or newer, 64 bits

[Release Notes](#)

(/wiki/File:ESP32-S3-AMOLED-1.91-Ar-software-01.png)

- Run the installer and install all by default

Install ESP32 Development Board

- Before using ESP32-related motherboards with the Arduino IDE, you must first install the software package for the **esp32 by Espressif Systems** development board
- According to **board installation requirement**, it is generally recommended to use **Install Online**. If online installation fails, use **Install Offline**.
- For the installation tutorial, please refer to Arduino board manager tutorial (https://www.waveshare.com/wiki/Arduino_Board_Managers_Tutorial)

- ESP32-S3-LCD-2 required development board installation description

Board name	Board installation requirement	Version number requirement
esp32 by Espressif Systems	"Install Offline" / "Install Online"	≥3.0.0

Install Library

- When installing Arduino libraries, there are usually two ways to choose from: **Install online** and **Install offline**. **If the library installation requires offline installation, you must use the provided library file**

For most libraries, users can easily search and install them through the online library manager of the Arduino software. However, some open-source libraries or custom libraries are not synchronized to the Arduino Library Manager, so they cannot be acquired through online searches. In this case, users can only manually install these libraries offline.

- For library installation tutorial, please refer to Arduino library manager tutorial (https://www.waveshare.com/wiki/Arduino_Library_Manager_Tutorial)
- **ESP32-S3-LCD-2 library file path:**

```
..\ESP32-S3-LCD-2-Demo\Arduino\libraries
```

■ ESP32-S3-LCD-2 library file installation description

Library Name	Description	Version	Library Installation Requirement
lvgl	Graphical library	v8.4.0	"Install Online" (requires copying the demos folder to src)
GFX_Library_for_Arduino	LCD driver library	v1.5.0	"Install Online"
FastIMU	IUM driver library	v1.2.6	"Install Online"
OneButton	Button driver library	v2.6.1	"Install Online"

Run the First Arduino Demo

If you are just getting started with ESP32 and Arduino, and you don't know how to create, compile, flash, and run Arduino ESP32 programs, then please

[\[Expand\]](#)

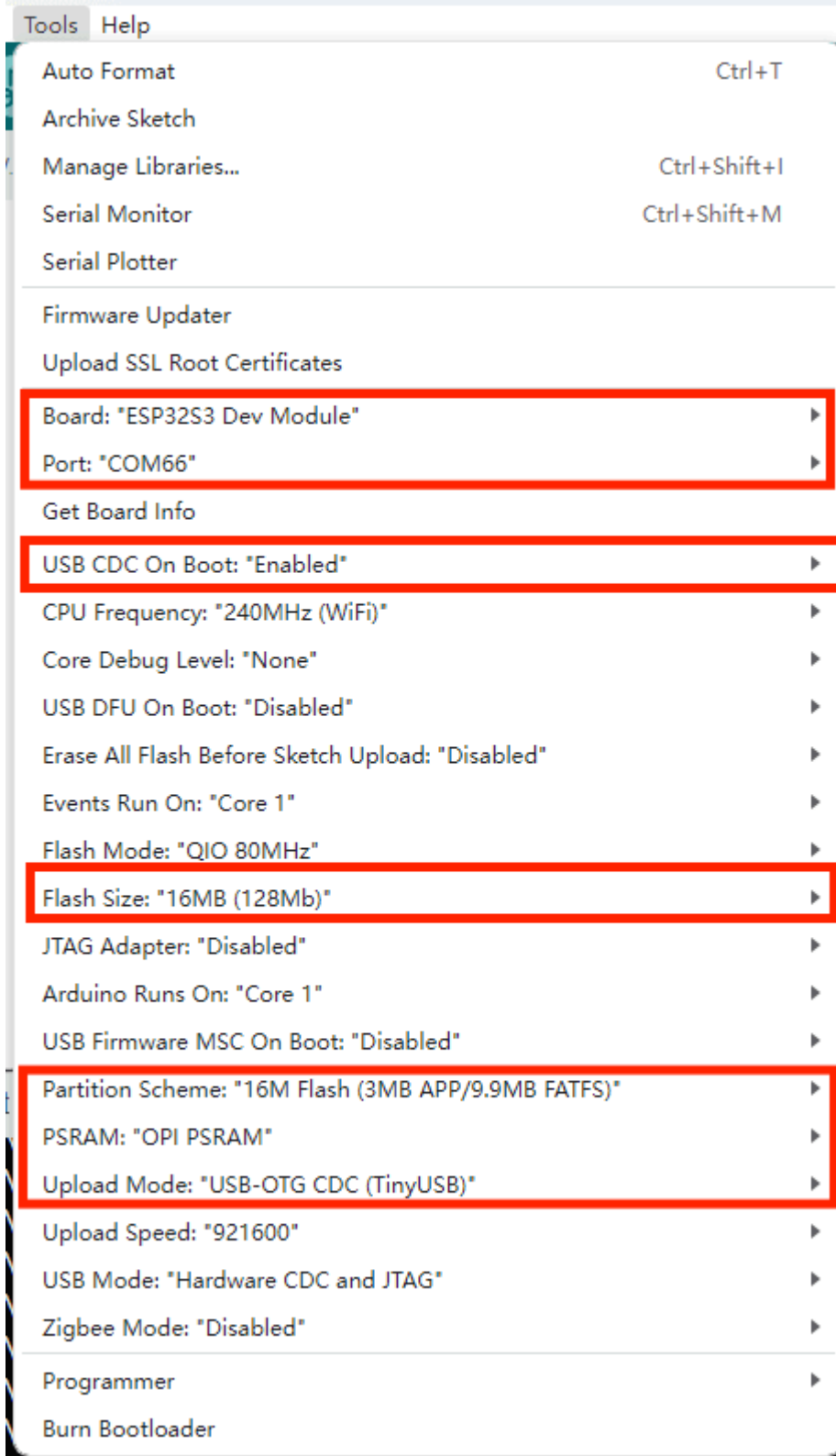
expand and take a look. Hope it can help you!

Demo

- ESP32-S3-LCD-2 demos

Demo	Basic Description	Dependency Library
01_factory	Comprehensive testing program	lvgl, GFX_Library_for_Arduino, FastIMU, OneButton
02_gfx_helloworld	HelloWorld is displayed on the screen	GFX_Library_for_Arduino
03_sd_card_test	Test TF card	--
04_qmi8658_output	Serial port prints QMI8658 data	FastIMU
05_lvgl_qmi8658	Use the LVGL library to display QMI8658 data	lvgl, FastIMU
06_lvgl_battery	Use the LVGL library to display the battery voltage	lvgl, GFX_Library_for_Arduino
07_lvgl_brightness	Use the LVGL library to control and display screen brightness	lvgl, GFX_Library_for_Arduino
08_lvgl_example	lvgl demos	lvgl, GFX_Library_for_Arduino
09_lvgl_camera	Use the LVGL library to display the camera images	lvgl, GFX_Library_for_Arduino
10_camera_web_server	Display the camera images on the web page	--

Arduino Project Parameter Setting



LCD-2-Demo-Arduino-Config.png)

(/wiki/File:ESP32-S3-Touch-

01_factory

【Demo description】

[\[Collapse\]](#)

This demo tests the functionality of the ESP32-S3-LCD-2 onboard module, the information of each module will be displayed on the screen, and the user can switch between pages through the touch screen

【Hardware connection】

- Connect the board to the computer
- Plug the OV5640 camera into the 24Pin socket on the board (if it is not available, it will not affect the use of other functions)
- Insert the TF card into the card slot (if it is not available, it will not affect the use of other functions)



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-0.jpg)

【Code analysis】

■ Hardware initialization

```
Serial.begin(115200);
bsp_i2c_init();          //Initialize I2C bus, commonly used to connect sensors, di
splays, etc.
bsp_lv_port_init();      //Initialize the LVGL graphics library port to output grap
hics to display devices such as LCD
bsp_spi_init();          //Initialize SPI bus, commonly used for communication with
sensors, external storage, screens, etc.
```

■ LVGL graphics library configuration

```
bsp_lv_port_run();       //Run LVGL-related drivers or loops, such as refreshing th
e display, etc.
if (lvgl_lock(-1)) {     //Lock LVGL resources to ensure they do not conflict with
other tasks when updating the UI
    lvgl_ui_init();      //Initialize UI, possibly including layout and control set
tings
    lvgl_unlock();       //Unlock LVGL, allowing other tasks to access LVGL
}
```

■ Peripheral initialization

```

app_qmi8658_init();           //Initialize QMI8658 sensor (such as accelerometer, gyroscope, etc.)
app_system_init();           //Perform system-level initialization, such as clock, memory, reset, etc
app_camera_init();           //Initialize the camera module, prepare for image capture
app_wifi_init(sta_ssid, sta_pass); //Initialize Wi-Fi module, connect to specified Wi-Fi network

```

■ Application layer tasks running

```

app_qmi8658_run();           //Start QMI8658 sensor-related tasks, such as data acquisition, processing, etc.
app_system_run();           //Start system tasks to ensure that system functions are running properly
app_camera_run();           //Start camera-related tasks, usually to obtain image data and process or transmit it
app_wifi_run();             //Start Wi-Fi related tasks, such as connection management, data transmission, etc.

```

【Result demonstration】

- Click the BOOT button to be down, double click the BOOT button to be up, and long press the BOOT button to OK
- System interface



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-1.jpg)

- QMI8658 interface, where you can see the acceleration and angular velocity data of the acquired X, Y, and Z axes



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-2.jpg)

- Camera interface, where you can see the images captured by the camera



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-3.jpg)

- WiFi interface, showing the WiFi name and password to be connected, IP will be displayed after successful connection



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-4.jpg)

02_gfx_helloworld

【Demo description】

[Collapse]

This demo demonstrates the ESP32-S3-LCD-2 using the GFX_Library_for_Arduino library to drive the screen and display HelloWorld on the screen.

【Hardware connection】

-
- Connect the board to the computer

【Code analysis】

-
- Create an object bus of the Arduino_ESP32SPI class to configure the SPI bus GPIO, and create an object gfx of the Arduino_ST7789 class to drive the ST7789 display

```
Arduino_DataBus *bus = new Arduino_ESP32SPI(
    EXAMPLE_PIN_NUM_LCD_DC /* DC */, EXAMPLE_PIN_NUM_LCD_CS /* CS */,
    EXAMPLE_PIN_NUM_LCD_SCLK /* SCK */, EXAMPLE_PIN_NUM_LCD_MOSI /* MOSI */, EXAMPLE_
PIN_NUM_LCD_MISO /* MISO */);

/* More display class: https://github.com/moononournation/Arduino\_GFX/wiki/Display-
Class */
Arduino_GFX *gfx = new Arduino_ST7789(
    bus, EXAMPLE_PIN_NUM_LCD_RST /* RST */, EXAMPLE_LCD_ROTATION /* rotation */, true
/* IPS */,
    EXAMPLE_LCD_H_RES /* width */, EXAMPLE_LCD_V_RES /* height */);
```

【Result demonstration】



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-5.jpg)

03_sd_card_test

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to test the read and write functions of the TF card

【Hardware connection】

- Connect the board to the computer
- Insert the TF card into the card slot (**TF card needs to be formatted as FAT32**)

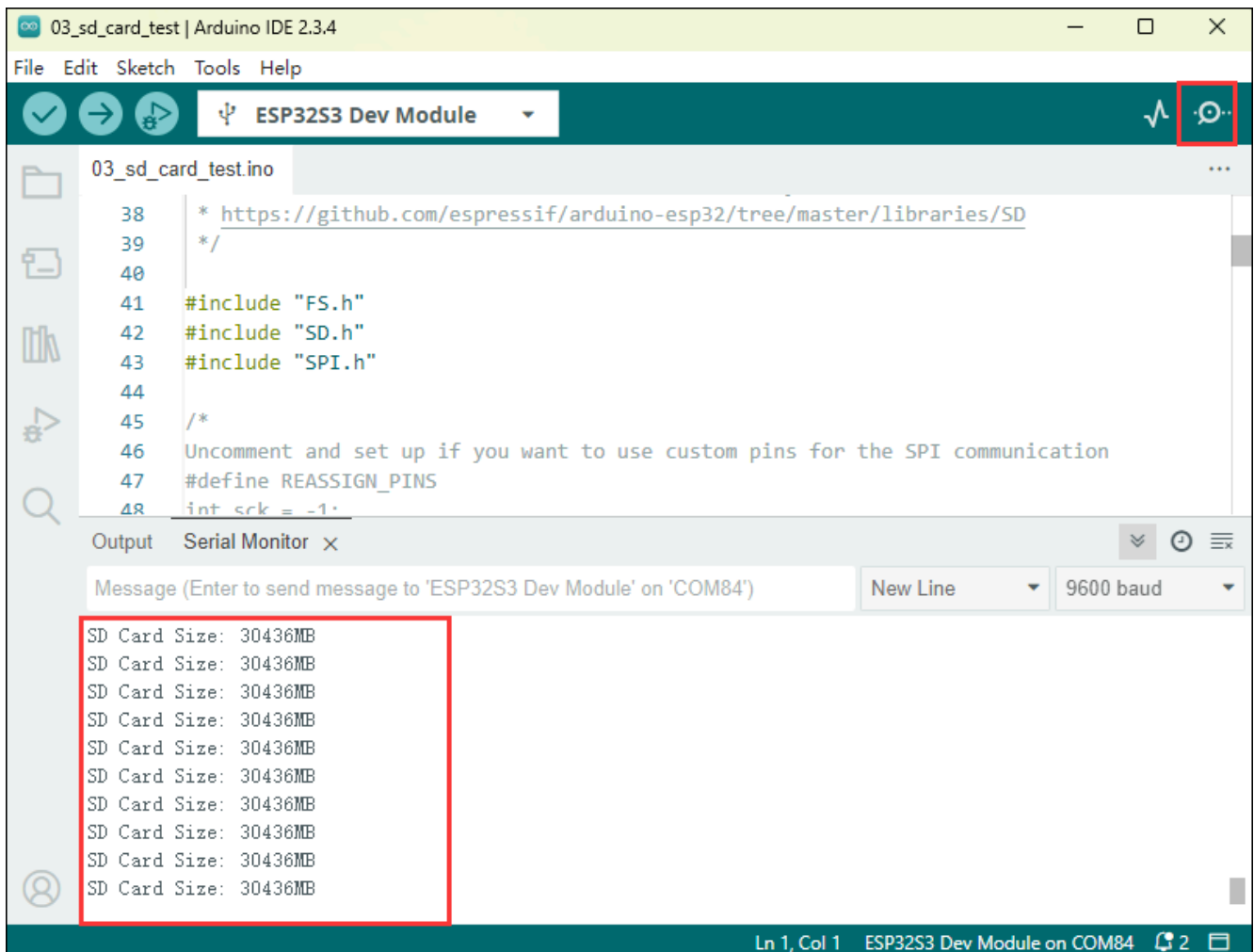
【Code analysis】

- SPI interface initialization and TF card initialization

```
SPI.begin(sck, miso, mosi, cs);  
if (!SD.begin(cs)) {  
    Serial.println("Card Mount Failed");  
    return;  
}
```

【Result demonstration】

- Open the serial port monitor



(/wiki/File:ESP32-S3-LCD-2-Demo-6.png)

- Insert the TF card into the computer, and you can find two more files: test.txt and foo.txt. The content of the foo.txt is Hello World!, and the content of the test.txt is empty

 foo.txt

 test.txt

(/wiki/File:ESP32-S3-LCD-2-Demo-7.png)

04_qmi8658_output

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to acquire QMI8658 data and print using serial port

【Hardware connection】

- Connect the board to the computer

【Code analysis】

- Initialize QMI8658

```
int err = IMU.init(calib, IMU_ADDRESS);
if (err != 0) {
    Serial.print("Error initializing IMU: ");
    Serial.println(err);
    while (true) {
        ;
    }
}
```

【Result demonstration】

- Open the serial port monitor to see the printed x, y, z axis accel and gyro as well as the qmi8658 internal temperature sensor data

05_lvgl_qmi8658

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to acquire QMI8658 data and display it using the lvgl library

【Hardware connection】

- Connect the board to the computer

【Code analysis】

- Initialize the UI and create a 100ms timer to obtain data from QMI8658

```
lvgl_qmi8658_ui_init(lv_scr_act());
```

【Result demonstration】

(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-9.jpg)

06_lvgl_battery

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to display battery voltage and ADC values on the screen using the lvgl library

【Hardware connection】

- Connect the board to the computer
- Insert the battery into the battery holder

【Code analysis】

- Initialize the UI and create a 1000ms timer to obtain the data of the adc and convert the data into a voltage

```
lvgl_battery_ui_init(lv_scr_act());
```

【Result demonstration】



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-10.jpg)

07_lvgl_brightness

【Demo description】

[Collapse]

This demo demonstrates how to use ESP32-S3-LCD-2 to display the screen brightness on the screen using the lvgl library, and the screen brightness can be controlled through the slider

【Hardware connection】

- Connect the board to the computer

【Code analysis】

- Initialize the UI, and create a slider value change callback, when the value of the slider changes, modify the screen brightness

```
lvgl_brightness_ui_init(lv_scr_act());
```

【Result demonstration】



2-Demo-11.jpg)

(/wiki/File:400px-ESP32-S3-LCD-

08_lvgl_example

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to run the lvgl demos

【Hardware connection】

- Connect the board to the computer

【Precautions】

- If the lvgl library is installed online, you need to copy the demos folder to src

【Code analysis】

- Select the lvgl demos to run

```
lv_demo_widgets();  
// lv_demo_benchmark();  
// lv_demo_keypad_encoder();  
// lv_demo_music();  
// lv_demo_stress();
```

【Result demonstration】

(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-12.jpg)

09_lvgl_camera

【Demo description】

[\[Collapse\]](#)

This demo uses the LVGL library to display images captured from the camera on the ESP32-S3-LCD-2 screen

【Hardware connection】

- Connect the board to the computer
- Insert the TF card into the card slot

【Code analysis】

- Create a task specifically for acquiring camera images

```
xTaskCreatePinnedToCore(camera_task, "camera_task_task", 1024 * 3, NULL, 1, NULL, 0);
```

- Obtain camera images and update display

```
camera_fb_t *pic;
lv_img_dsc_t img_dsc;
img_dsc.header.always_zero = 0;
img_dsc.header.w = 480;
img_dsc.header.h = 320;
img_dsc.data_size = 320 * 480 * 2;
img_dsc.header.cf = LV_IMG_CF_TRUE_COLOR;
img_dsc.data = NULL;
// lv_img_set_src(img_camera, &pic);
while (1)
{
    pic = esp_camera_fb_get();

    if (NULL != pic)
    {
        img_dsc.data = pic->buf;
        if (lvgl_lock(-1))
        {
            lv_img_set_src(img_camera, &img_dsc);
            lvgl_unlock();
        }
    }
    esp_camera_fb_return(pic);
    vTaskDelay(pdMS_TO_TICKS(1));
}
```

【Result demonstration】

2-Demo-13.jpg)

10_camera_web

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to drive the camera. After connecting to WiFi, the program will create a web server, and users can access it by entering the device's IP address in the browser. Web pages can display images from cameras and support settings such as resolution and mirroring

【Hardware connection】

-
- Connect the board to the computer
 - Plug the OV5640 camera into the 24Pin socket on the board

【Code analysis】

-
- `char sta_ssid[] = "waveshare";` : The name of the WiFi to be connected
 - `char sta_pass[] = "12345678";` : The WiFi password to be connected

【Result demonstration】

-
- Open the serial port terminal after connecting to WiFi to see the IP address

(/wiki/File:ESP32-S3-LCD-2-Demo-14.png)

- Use the browser to open the IP address for serial port printing
- Click Start Stream to see the image of the camera

(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-15.png)

Working with ESP-IDF

This chapter introduces setting up the ESP-IDF environment setup, including the installation of Visual Studio and the Espressif IDF plugin, program compilation, downloading, and testing of demos, to assist users in mastering the development board and facilitating secondary development.

Environment Setup

Download and install Visual Studio

- Open the download page of VScode official website (<https://code.visualstudio.com/download>), choose the corresponding system and system bit to download

Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



↓ Windows

Windows 10, 11



↓ .deb

Debian, Ubuntu

↓ .rpm

Red Hat, Fedora, SUSE



↓ Mac

macOS 10.15+

User Installer [x64](#) [Arm64](#)

System [x64](#) [Arm64](#)

Installer [x64](#) [Arm64](#)

.zip [x64](#) [Arm64](#)

CLI [x64](#) [Arm64](#)

.deb [x64](#) [Arm32](#) [Arm64](#)

.rpm [x64](#) [Arm32](#) [Arm64](#)

.tar.gz [x64](#) [Arm32](#) [Arm64](#)

Snap [Snap Store](#)

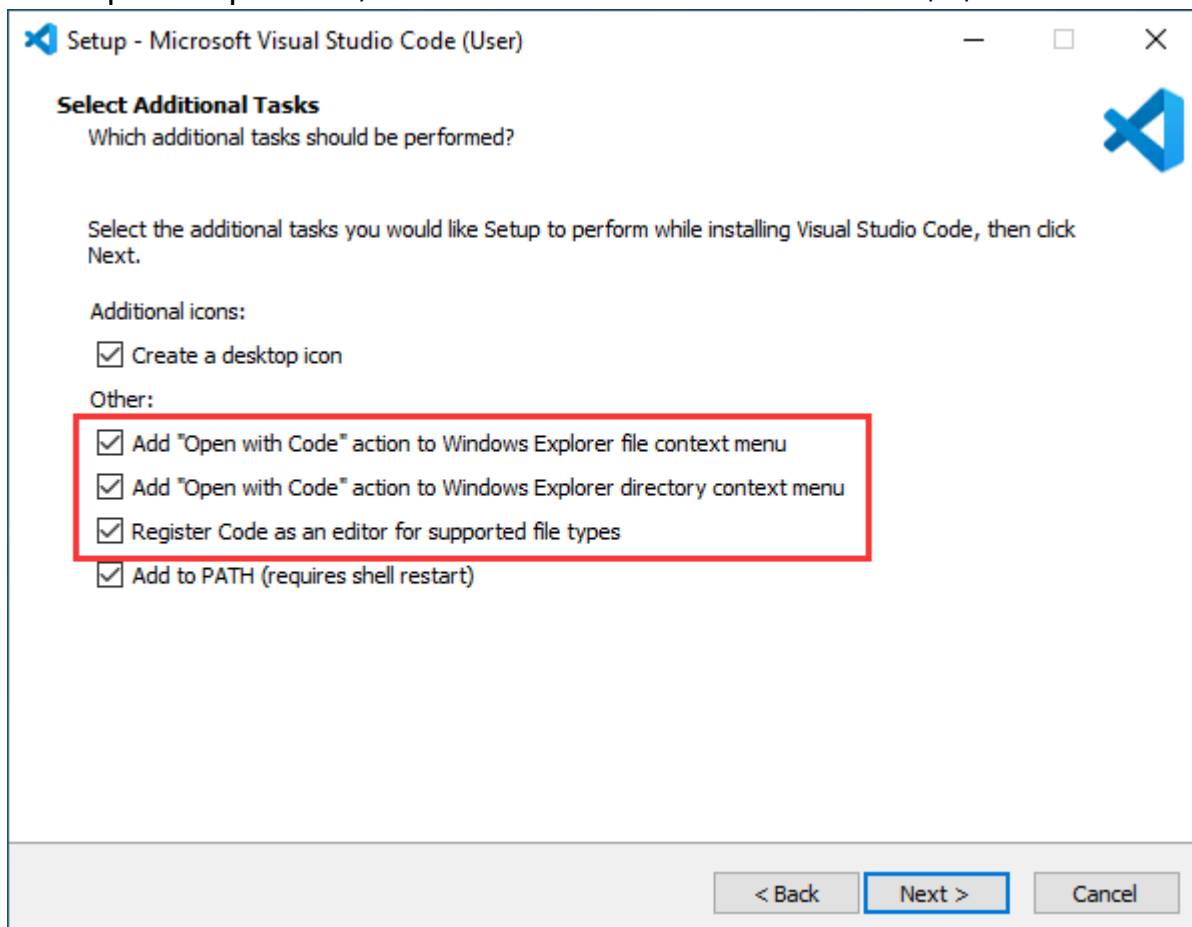
CLI [x64](#) [Arm32](#) [Arm64](#)

.zip [Intel chip](#) [Apple silicon](#) [Universal](#)

CLI [Intel chip](#) [Apple silicon](#)

(/wiki/File:ESP32-S3-AMOLED-1.91-VScode-01.png)

- After running the installation package, the rest can be installed by default, but here for the subsequent experience, it is recommended to check boxes 1, 2, and 3



(/wiki/File:ESP32-S3-AMOLED-1.91-VScode-02.png)

- After the first two items are enabled, you can open VSCode directly by right-clicking files or directories, which can improve the subsequent user experience
- After the third item is enabled, you can select VSCode directly when you choose how to open it

The environment setup is carried out on the Windows 10 system, Linux and Mac users can access ESP-IDF environment setup (<https://docs.espressif.com/projects/esp-idf/en/v5.1.4/esp32s3/get-started/windows-setup.html>) for reference

Install Espressif IDF Plugin

- It is generally recommended to use **Install Online**. If online installation fails due to network factor, use **Install Offline**
- For more information about how to install the Espressif IDF plugin, see Install Espressif IDF Plugin (https://www.waveshare.com/wiki/Install_Espressif_IDF_Plugin_Tutorial)

Run the First ESP-IDF Demo

If you are just getting started with ESP32 and ESP-IDF, and you don't know how to create, compile, flash, and run ESP-IDF ESP32 programs, then please expand and take a look. Hope it can help you!

[\[Expand\]](#)

Demo

- ESP32-S3-LCD-2 demos

Demo	Basic Description
sd_card_test	Test TF card
lvgl_example	Display lvgl demos
lvgl_qmi8658	Use lvgl library to display qmi8658 data
lvgl_camera	Use lvgl library to display camera images
lvgl_battery	Use lvgl library to display battery voltage
lvgl_brightness	Use lvgl library to control and display screen brightness

sd_card_test

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to test the read and write functions of the TF card.

【Hardware connection】

- Connect the board to the computer
- Insert the TF card into the card slot

【Code analysis】

■ SPI initialization

```
sdmmc_host_t host = SDSPI_HOST_DEFAULT();

spi_bus_config_t bus_cfg = {
    .mosi_io_num = PIN_NUM_MOSI,
    .miso_io_num = PIN_NUM_MISO,
    .sclk_io_num = PIN_NUM_CLK,
    .quadwp_io_num = -1,
    .quadhd_io_num = -1,
    .max_transfer_sz = 4000,
};
ret = spi_bus_initialize(host.slot, &bus_cfg, SDSPI_DEFAULT_DMA);
if (ret != ESP_OK) {
    ESP_LOGE(TAG, "Failed to initialize bus.");
    return;
}
```

■ TF card initialization and mounting

```
sdspi_device_config_t slot_config = SDSPI_DEVICE_CONFIG_DEFAULT();
slot_config.gpio_cs = PIN_NUM_CS;
slot_config.host_id = host.slot;

ESP_LOGI(TAG, "Mounting filesystem");
ret = esp_vfs_fat_sdspi_mount(mount_point, &host, &slot_config, &mount_config,
&card);
```

【Result demonstration】

■ Serial port monitor

(/wiki/File:600px-ESP32-S3-LCD-2-Demo-16.png)

- Insert the TF card into the computer, and you can find three more files: test.txt, FOO.TXT and NIHAO.TXT. Among them, the content of the FOO.TXT is Hello SD!, the content of the NIHAO.TXT is Hello SD!, and the content of the test.txt is empty

 FOO.TXT

 NIHAO.TXT

(/wiki/File:ESP32-S3-LCD-2-Demo-17.png)

 test.txt

【Precautions】

- TF card needs to be formatted as FAT32

lvgl_example

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to run the lvgl demos

【Hardware connection】

- Connect the board to the computer

【Code analysis】

- Initialization

```
lv_init();
display_init();
lv_port_disp_init();
lvgl_tick_timer_init(EXAMPLE_LVGL_TICK_PERIOD_MS);
bsp_brightness_init();
bsp_brightness_set_level(80);
```

- Select the lvgl demos to run

```
lv_demo_widgets();
// lv_demo_benchmark();
// lv_demo_keypad_encoder();
// lv_demo_music();
// lv_demo_stress();
```

【Result demonstration】



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-3.jpg)

lvgl_qmi8658

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to acquire QMI8658 data and display it using the lvgl library

【Hardware connection】

- Connect the board to the computer

【Code analysis】

- Initialize the UI and create a 100ms timer to obtain data from QMI8658

```
lvgl_qmi8658_ui_init(lv_scr_act());
```

【Result demonstration】



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-18.jpg)

lvgl_camera

【Demo description】

[\[Collapse\]](#)

This demo uses the LVGL library to display images captured from the camera on the ESP32-S3-LCD-2 screen

【Hardware connection】

- Connect the board to the computer
- Insert the TF card into the card slot

【Code analysis】

- Create a task specifically for acquiring camera images

```
xTaskCreatePinnedToCore(camera_task, "camera_task_task", 1024 * 3, NULL, 1, NULL, 0);
```

- Obtain camera images and update display

```
camera_fb_t *pic;
lv_img_dsc_t img_dsc;
img_dsc.header.always_zero = 0;
img_dsc.header.w = 480;
img_dsc.header.h = 320;
img_dsc.data_size = 320 * 480 * 2;
img_dsc.header.cf = LV_IMG_CF_TRUE_COLOR;
img_dsc.data = NULL;
// lv_img_set_src(img_camera, &pic);
while (1)
{
    pic = esp_camera_fb_get();

    if (NULL != pic)
    {
        img_dsc.data = pic->buf;
        if (lvgl_lock(-1))
        {
            lv_img_set_src(img_camera, &img_dsc);
            lvgl_unlock();
        }
    }
    esp_camera_fb_return(pic);
    vTaskDelay(pdMS_TO_TICKS(1));
}
```

【Result demonstration】

(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-13.jpg)

lvgl_battery

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to display battery voltage and ADC values on the screen using the lvgl library

【Hardware connection】

- Connect the board to the computer
- Insert the battery into the battery holder

【Code analysis】

- Initialize the UI and create a 1000ms timer to obtain the data of the adc and convert the data into a voltage

```
lvgl_battery_ui_init(lv_scr_act());
```

【Result demonstration】



(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-10.jpg)

lvgl_brightness

【Demo description】

[\[Collapse\]](#)

This demo demonstrates how to use ESP32-S3-LCD-2 to display the screen brightness on the screen using the lvgl library, and the screen brightness can be controlled through the slider

【Hardware connection】

- Connect the board to the computer

【Code analysis】

- Initialize the UI, and create a slider value change callback, when the value of the slider changes, modify the screen brightness

```
lvgl_brightness_ui_init(lv_scr_act());
```

【Result demonstration】

(/wiki/File:400px-ESP32-S3-LCD-

2-Demo-19.jpg)

Flash Firmware Flashing and Erasing

- The current demo provides test firmware, which can be used to test whether the onboard device functions properly by directly flashing the test firmware
- bin file path:

```
..\ESP32-S3-LCD-2-Demo\Firmware
```

Flash firmware flashing and erasing (https://www.waveshare.com/wiki/Flash_Firmware_Flashing_and_Erasing) for reference

Resources

Schematic Diagram

- ESP32-S3-LCD-2 Schematic diagram (<https://files.waveshare.com/wiki/ESP32-S3-LCD-2/ESP32-S3-Touch-LCD-2-SchDoc.pdf>)

Demo

- ESP32-S3-LCD-2 Demo (<https://files.waveshare.com/wiki/ESP32-S3-LCD-2/ESP32-S3-LCD-2-Demo.zip>)

Project Document

- ESP32-S3-LCD-2 2D/3D file (<https://files.waveshare.com/wiki/ESP32-S3-LCD-2/ESP32-S3-LCD-2-2Dand3D.rar>)

Datasheets

ESP32-S3

- ESP32-S3 Technical Reference Manual (https://files.waveshare.com/wiki/common/Esp32-s3_technical_reference_manual_en.pdf)
- ESP32-S3 Series Datasheet (https://files.waveshare.com/wiki/common/Esp32-s3_datasheet_en.pdf)

Software Tools

Arduino

- Arduino IDE Official download link (<https://www.arduino.cc/en/software/>)
- ESP32-Arduino official documentation (<https://docs.espressif.com/projects/arduino-esp32/en/latest/index.html>)

- Arduino-ESP32 offline component package (https://drive.google.com/drive/folders/1Pcs_A4FKWvdSHnz9IEBYqOpr-noTMblv?usp=sharing)

VSCode

- VSCode official website (<https://code.visualstudio.com/download>)

Firmware Flashing Tool

- Flash_download_tool (https://dl.espressif.com/public/flash_download_tool.zip)

Other Resource Links

- ESP32-Arduino official documentation (<https://docs.espressif.com/projects/arduino-esp32/en/latest/index.html>)
- LVGL official documentation (<https://docs.lvgl.io/master/intro/introduction/index.html>)

FAQ

Question: What should I do if the Arduio demo doesn't display after downloading?

Answer:

- Refer to Arduino Project Parameter Setting (<https://www.waveshare.com/wiki/ESP32-S3-Touch-LCD-2#Demo>).

Question: After the module downloads the demo and re-downloads it, why sometimes it can't connect to the serial port or the flashing fails?

Answer:

- Long press the BOOT button, plug in the USB at the same time, then release the BOOT button, at this time the module can enter the download mode, which can solve most of the problems that can not be downloaded.

Question: Why does the module keep resetting and flicker when viewed the recognition status from the device manager?

Answer:

- It may be due to Flash blank and the USB port is not stable, you can long-press the BOOT button, press RESET at the same time, and then release RESET, and then release the BOOT button, at this time the module can enter the download mode to flash the firmware (demo) to solve the situation.

Question: How to deal with the first compilation of the program being extremely slow?

Answer:

- It's normal for the first compilation to be slow, just be patient

Question: How to handle the display "waiting for download..." on the serial port after successfully ESP-IDF flashing?

Answer:

- If there is a reset button on the development board, press the reset button; if there is no reset button, please power it on again

Question: What should I do if I can't find the AppData folder?

Answer:

- Some AppData folders are hidden by default and can be set to show.
- English system: Explorer->View->Check "Hidden items"
- Chinese system: File Explorer -> View -> Display -> Check "Hidden Items"

Question: How do I check the COM port I use?

Answer:

- Windows system:

①View through Device Manager: Press the Windows + R keys to open the "Run" dialog box; input devmgmt.msc and press Enter to open the Device Manager; expand the "Ports (COM and LPT)" section, where all COM ports and their current statuses will be listed.

②Use the command prompt to view: Open the Command Prompt (CMD), enter the "mode" command, which will display status information for all COM ports.

③Check hardware connections: If you have already connected external devices to the COM port, the device usually occupies a port number, which can be determined by checking the connected hardware.

- Linux system:

①Use the dmesg command to view: Open the terminal.

①Use the ls command to view: Enter `ls /dev/ttyS*` or `ls /dev/ttyUSB*` to list all serial port devices.

③Use the setserial command to view: Enter `setserial -g /dev/ttyS*` to view the configuration information of all serial port devices.

Question: Why does the program flashing fail when using a MAC device?

Answer:

- Install MAC Driver (https://files.waveshare.com/wiki/common/CH34XSER_MAC.7z) and flash again.

Question: How to use SquareLine Studio to design interfaces?

Answer:

- Please refer to SquareLine Studio tutorial (https://www.waveshare.com/wiki/Waveshare_SquareLine_Studio)

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.

Working Time: 9 AM - 6 PM GMT+8
(Monday to Friday)

Submit Now (<https://service.waveshare.com/>)

